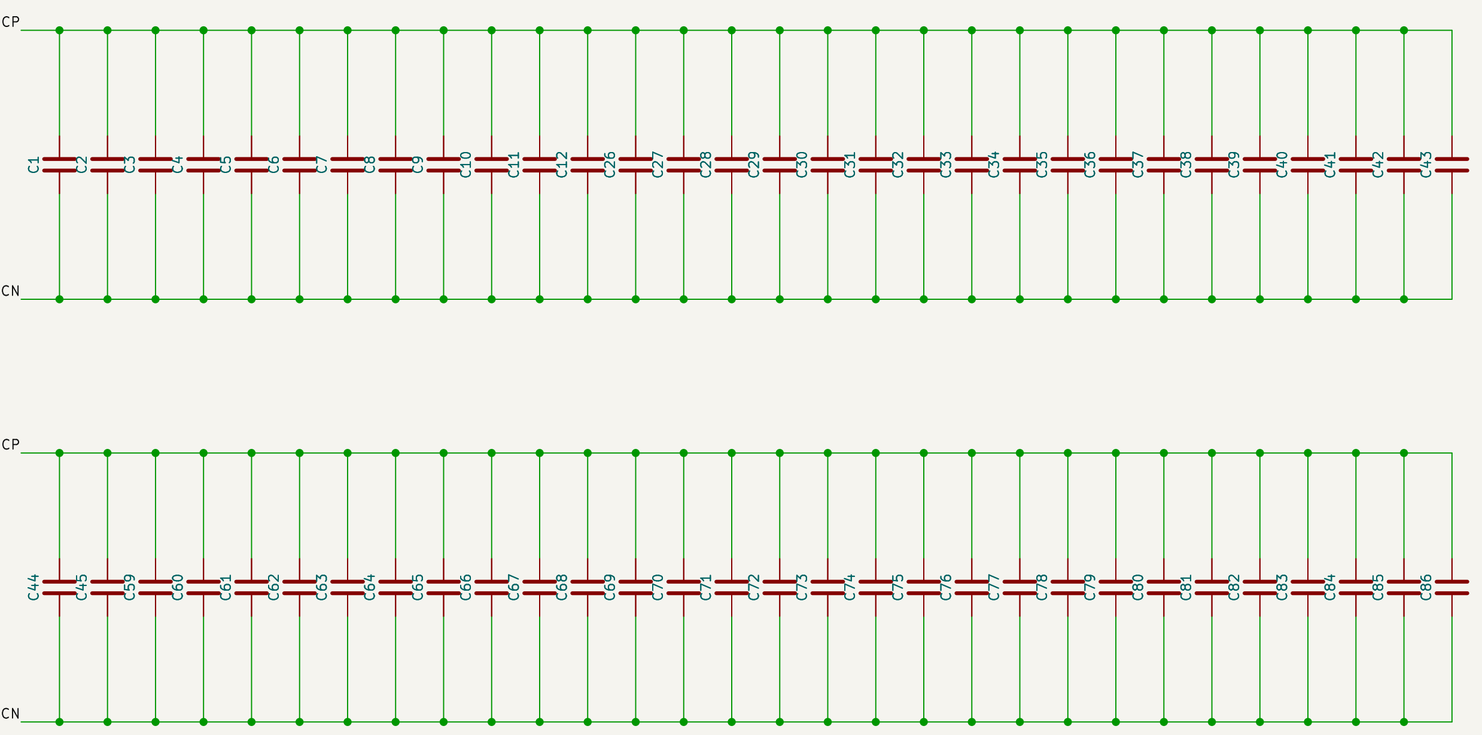
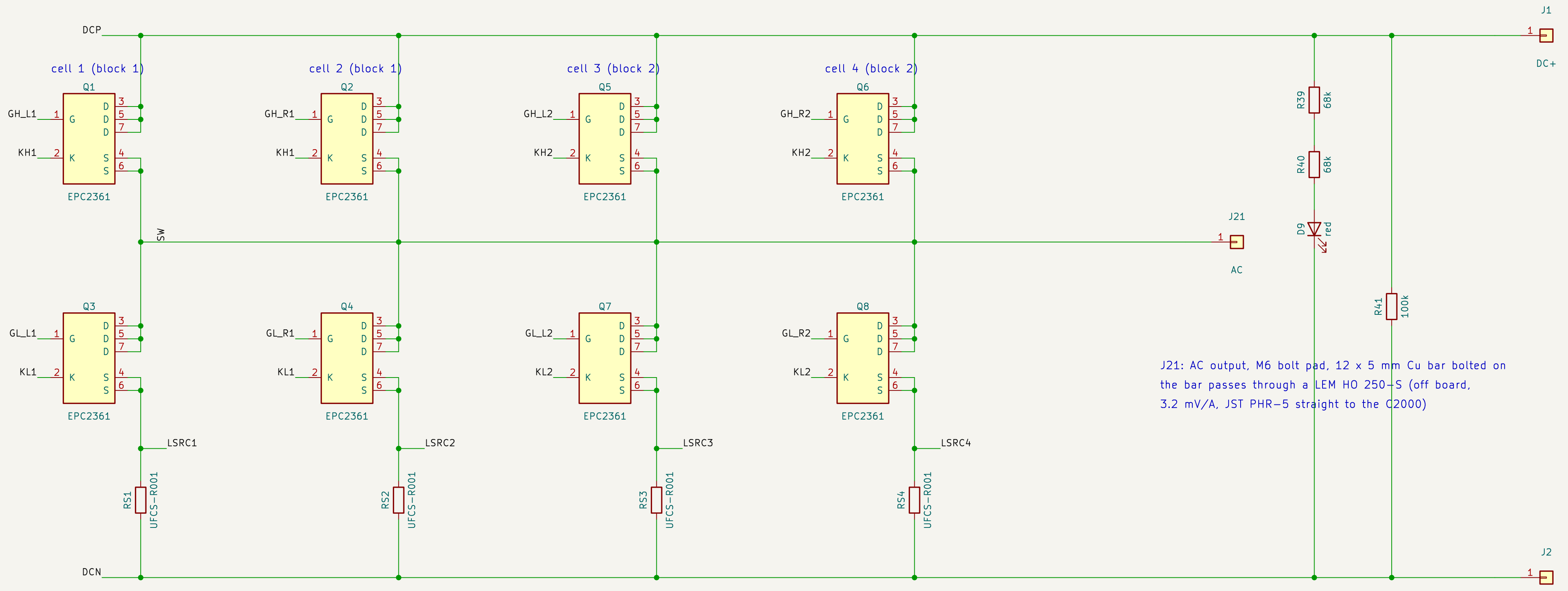
## POWER STAGE AND DC LINK  (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor + UFCS shunt in the LS source; AC of all four cells joined on the PCB)


DCP
DCN
DCP
DCN

DC link, 60 capacitors: 40 x 1uF 100V X7S,  20 x 22uF 100V X7S 2220  (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)


DCP
cell 1 (block 1)
cell 2 (block 1)
cell 3 (block 2)
cell 4 (block 2)
Q1
Q2
Q5
Q6
EPC2361
GH_L1
GH_R1
GH_L2
GH_R2
KH1
KH2
SW
J21
AC
Q3
Q4
Q7
Q8
GL_L1
GL_R1
GL_L2
GL_R2
KL1
KL2
LSRC1
LSRC2
LSRC3
LSRC4
RS1
RS2
RS3
RS4
UFCS-R001
R39
68k
R40
68k
D9
red
R41
100k
J1
DC+
J2
DC-
DCN

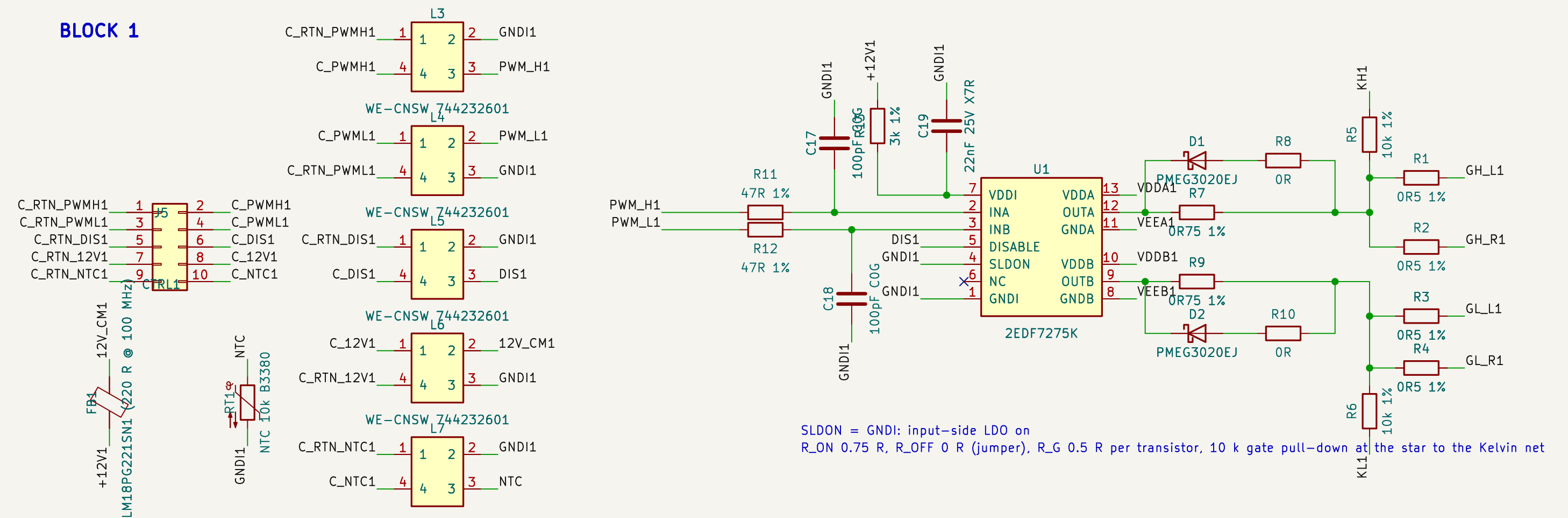
RS1–RS4: PMK UFCS–R001 1 mOhm coaxial shunt, IN = LS source (LSRCx), SHLD = DC–

J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on
the bar passes through a LEM HO 250–S (off board,
3.2 mV/A, JST PHR–5 straight to the C2000)

DC link present LED, R41 100 k bleeder

## GATE DRIVER BLOCKS: control header with common–mode chokes, 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

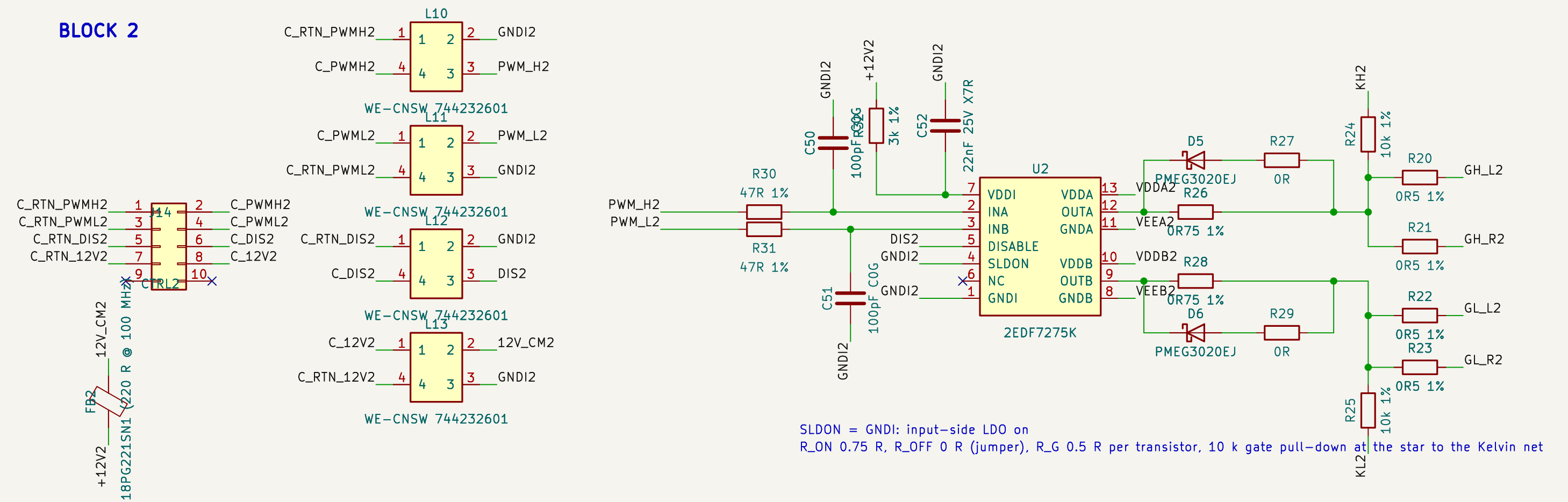
**BLOCK 1**


L3
C_RTN_PWMH1
C_PWMH1
GNDI1
PWM_H1
WE-CNSW 744232601
L4
C_PWML1
C_RTN_PWML1
PWM_L1
GNDI1
L5
C_RTN_DIS1
C_DIS1
DIS1
L6
C_12V1
C_RTN_12V1
12V_CM1
L7
C_RTN_NTC1
C_NTC1
NTC
J5
C_RTN_PWMH1
C_RTN_PWML1
C_RTN_DIS1
C_RTN_12V1
C_RTN_NTC1
C_PWMH1
C_PWML1
C_DIS1
C_12V1
C_NTC1
FB1
BLM18PG221SN1
220 R @ 100 MHz
+12V1
RT1
NTC 10k B3380
C17
100pF C0G
R10
3k 1%
C19
22nF 25V X7R
+12V1
R11
47R 1%
R12
47R 1%
PWM_H1
PWM_L1
C18
100pF C0G
U1
2EDF7275K
VDDI
INA
INB
DISABLE
SLDON
NC
GNDI
VDDA
OUTA
GNDA
VDDB
OUTB
GNDB
VDDA1
VEEA1
VDDB1
VEEB1
D1
D2
PMEG3020EJ
R7
R9
0R75 1%
R8
0R
R5
10k 1%
R6
10k 1%
R1
R2
R3
R4
0R5 1%
GH_L1
GH_R1
GL_L1
GL_R1
KH1
KL1

NTC on the control side (board temperature at the driver)

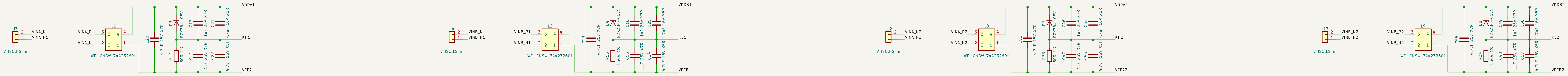
SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

**BLOCK 2**


L10
C_RTN_PWMH2
C_PWMH2
GNDI2
PWM_H2
WE-CNSW 744232601
C_PWML2
C_RTN_PWML2
PWM_L2
C_RTN_DIS2
C_DIS2
DIS2
C_12V2
C_RTN_12V2
12V_CM2
FB2
BLM18PG221SN1
220 R @ 100 MHz
+12V2
R30
47R 1%
R31
47R 1%
PWM_H2
PWM_L2
C50
C51
100pF C0G
C52
22nF 25V X7R
U2
2EDF7275K
VDDA2
VEEA2
VDDB2
VEEB2
D5
D6
PMEG3020EJ
R26
R28
0R75 1%
R27
R29
0R
R24
R25
10k 1%
R20
R21
R22
R23
0R5 1%
GH_L2
GH_R2
GL_L2
GL_R2
KH2
KL2

SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

## GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) –> WE–CNSW –> 5.1 V Zener VDD–S, 150 R bias S–VEE: V_GS,on = +5.1 V, V_GS,off = –(V_ISO – 5.1 V)


J3
VINA_N1
VINA_P1
V_ISO,HS in
L1
WE-CNSW 744232601
C20
4.7uF 25V X7R
D3
BZX384-C5V1
R14
150R 1%
C13
C14
1uF 25V X7R
C21
C22
4.7uF 16V X5R
VDDA1
KH1
VEEA1
J4
VINB_N1
VINB_P1
V_ISO,LS in
L2
C23
D4
R15
C16
C15
C25
C24
VDDB1
KL1
VEEB1
J12
VINA_N2
VINA_P2
L8
VDDA2
KH2
VEEA2
J13
VINB_N2
VINB_P2
L9
VDDB2
KL2
VEEB2

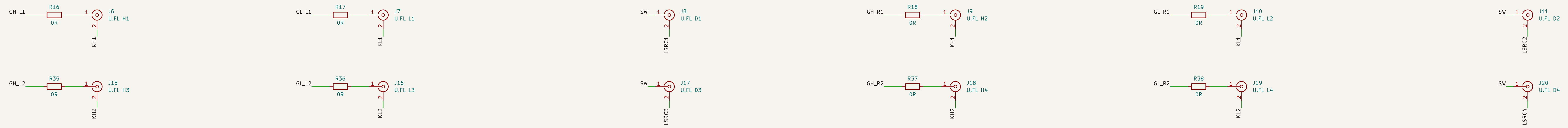
J3 / L1: HS block 1

J4 / L2: LS block 1

J12 / L8: HS block 2

J13 / L9: LS block 2

## U.FL MEASUREMENT PORTS (bottom side, at each cell): V_GS HS / LS (0 R, shield = Kelvin), V_DS LS (centre pin directly on the switch node, 1 MOhm scope input only – never 50 R at 75 V; shield = LS source LSRCx)


GH_L1
R16
0R
J6
U.FL H1
KH1
GL_L1
R17
0R
J7
U.FL L1
KL1
SW
J8
U.FL D1
LSRC1
GH_R1
R18
0R
J9
U.FL H2
KH1
GL_R1
R19
0R
J10
U.FL L2
KL1
SW
J11
U.FL D2
LSRC2
GH_L2
R35
0R
J15
U.FL H3
KH2
GL_L2
R36
0R
J16
U.FL L3
KL2
SW
J17
U.FL D3
LSRC3
GH_R2
R37
0R
J18
U.FL H4
KH2
GL_R2
R38
0R
J19
U.FL L4
KL2
SW
J20
U.FL D4
LSRC4

## MOUNTING: H1–H4 board corners (insulated); no heatsink (double–pulse board)

H1
H2
H3
H4
M2.5
M2.5
M2.5
M2.5

Nets: DCP = DC+, DCN = DC–, SW = joined switch node of all four cells.  KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361); they meet the power source only inside the package.

LSRC1–LSRC4: LS power source of each cell = UFCS IN; the shunt SHLD side is DC–.  The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC–.

Block suffix 1/2 on every driver–side net (GNDI1, VDDA2, ...).  Control side (CTRLx 2 x 5 header): PWM_H, PWM_L, DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE–CNSW common–mode choke per pair; 12 V then through a ferrite bead.  The control side is a top–layer–only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb–rules/PCB_rules_and_checklist.md.

No short–circuit protection on this board.  Closed–loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, UFCS shunt per cell, no heatsink, external isolated supplies
**KTH**
Sheet: /
File: spb–bb–revb–dpt.kicad_sch
***Title: SPB leg, Rev B, double–pulse board***
Size: A1  Date: 2026–10–14  **Rev: B**
KiCad E.D.A.  kicad–cli 7.0.10  Id: 1/1